

From Polygonal Quantum Control to an Emergent Local Metric

Finite-direction obstructions, nonorthogonal qutrit localization,
and a route to continuous predictive geometry

QBist Spacetime Reinforcement-Learning Project

August 11, 2026

Abstract

Can ordinary two-dimensional space arise as the geometry of goals for an agent controlling a quantum system whose Hilbert space has only three dimensions? Earlier exact constructions answer the cardinality question affirmatively: a qubit or qutrit has continuously many nonorthogonal states, and priced displacement instruments can make a finite goal set have exact Euclidean distances. Those constructions do not yet explain locality because they include an action tailored to every desired displacement.

This paper studies the stronger local problem. We construct a translation-covariant qutrit phase orbit and restrict the agent to one finite, goal-independent catalog of local binary Kraus instruments. We prove that retry instruments reduce exactly to a weighted word metric and that the large-scale unit ball is the convex hull of normalized primitive velocities. A fixed finite catalog therefore induces a polygonal Finsler norm, not a Euclidean norm. If its largest angular gap is $\Delta_{\max}$, the worst asymptotic excess is $\sec(\Delta_{\max}/2) - 1$; approximately uniform K -direction catalogs converge only as $O(K^{-2})$.

Deterministic inverse-design experiments confirm this obstruction beyond the fitting region. Held-out Euclidean relative RMSE falls from 30.03% with four directions to 3.85% with eight and 1.62% with 32. The 32-direction geometry has two-dimensional MDS stress 0.0289, but retains Schoenberg negative-eigenmass fraction 0.082, preventing an exact Euclidean interpretation. Sixteen nominal directions do no better than eight because their new moves are Bellman-dominated.

We then hide the initial position on a nine-state nonorthogonal qutrit phase grid and derive an exact Bayesian quantum filter for weak covariant measurements. In 60,000 episodes, informative measurements improve preparation-label navigation above the null value $1/9$, while known-start navigation is exact. Nevertheless, a sharp measurement identifies the initial label with probability only $1/3$ while preparing a present state that can be translated to the target with fidelity one. Historical localization and present-state control are inequivalent. Together the results identify a base action geometry and an epistemic quantum fiber, and motivate a next model with continuous qutrit controls, quadratic effort, and a learned predictive atlas.

Contents

1	Introduction	4
1.1	Space as a geometry of achievable goals	4
1.2	Low Hilbert dimension is not low goal cardinality	4
1.3	The locality problem	4
1.4	Contributions	4

2	Quantum and reinforcement-learning background	5
2.1	Quantum instruments	5
2.2	Bellman optimality	5
2.3	Sequence goals and memory	5
2.4	Metric and Euclidean tests	6
3	Translation-covariant qutrit controls	6
3.1	Equilateral phase encoding	6
3.2	Local projective geometry	6
3.3	Local Kraus attempts	7
4	The finite-direction theorem	7
4.1	Exact reduction to a word metric	7
4.2	Exact Euclidean rays	7
4.3	Polygonal stable norm	8
4.4	Angular error and exact examples	8
5	Control inverse design and evaluation	9
5.1	Repertoires	9
5.2	Targets, optimization, and held-out test	9
6	Control results	9
7	Hidden-start localization	11
7.1	Why geometry alone is insufficient	11
7.2	Weak covariant sensor	11
7.3	Exact Bayesian quantum filter	11
7.4	Two goals on the same history	12
7.5	Policies and controls	12
8	Localization results	13
8.1	Protocol and representative results	13
9	Joint geometric interpretation	13
9.1	A Finsler base	13
9.2	An epistemic quantum fiber	13
9.3	Goal relativity	15
10	Limitations	15
11	Continuous control and a predictive atlas	15
11.1	Replace the polygon by a disk	15
11.2	Quadratic effort and a metric tensor	16
11.3	Learn the predictive state	16
11.4	Genuine active localization	16
11.5	Base, fiber, and curvature tests	16
11.6	Falsifiers	16
12	Conclusion	17

A	Stable-norm duality and active actions	17
B	Statistical formulas	17
C	Reproducibility	18

1 Introduction

1.1 Space as a geometry of achievable goals

The working hypothesis is operational: an agent need not begin with physical space as a primitive. What it directly encounters is a collection of goals and interventions. Some goals require few reliable interventions; others require longer, riskier, or more information-intensive strategies. This induces a *hodological* geometry—a geometry of routes and goal difficulty—before coordinates are assigned.

Let z be an operational state, g a goal, and τ_g its first hitting time. For a policy π ,

$$d_\pi(z, g) = \mathbb{E}_\pi \left[\sum_{t=0}^{\tau_g-1} c(z_t, a_t) \mid z_0 = z \right]. \quad (1)$$

Optimizing gives $D(z, g) = \inf_\pi d_\pi(z, g)$. In general D can be directed, nonmetric, and high-dimensional. Ordinary space would be a special regime in which many place-like goals have nearly symmetric costs, admit a low-distortion common embedding, and support local homogeneous displacements. This uses the machinery of reinforcement learning (RL), partially observed control, and stochastic shortest paths [1, 2, 3, 4], but reverses the usual interpretation: values are evidence for whether space should emerge, not merely tools for navigation in a space assumed in advance.

1.2 Low Hilbert dimension is not low goal cardinality

A d -dimensional Hilbert space has at most d mutually orthogonal pure states but continuously many distinct density operators. Orthogonality limits perfect one-shot discrimination, not controlled-orbit size. Nine goals need not be nine basis states in a nine-dimensional position register. They can be nine nonorthogonal qutrit states or nine sequence-goal conditions.

This freedom has a danger: an unrestricted classical goal automaton can manufacture any finite graph while the quantum system remains idle. Physical Hilbert dimension, predictive-state dimension, and controller-memory dimension must be distinguished. Exact state preparation must also be distinguished from localization, because measurement can prepare an easy-to-control state while erasing its origin.

1.3 The locality problem

An earlier exact benchmark supplies one retry action for each source–goal difference δ , with success probability $p_\delta = 1/\|\delta\|_2$. Its expected attempt count is exactly $\|\delta\|_2$. This proves possibility, but explicitly puts Euclidean length into goal-sized macro-actions. The stronger question is:

Can a fixed, small, translation-covariant family of local qutrit interventions generate an ordinary metric by composition, including on directions and distances not used to tune it?

The answer is approximately yes and exactly no for fixed finite headings. The obstruction is geometric: additive local control with finitely many velocities has a polygonal unit ball.

1.4 Contributions

We provide:

1. an exact reduction from random-unitary retry instruments to weighted word metrics and a proof of their polygonal stable norms;

2. deterministic inverse design for four qutrit control stencils, tested on held-out displacements with Bellman, locality, MDS, and Schoenberg diagnostics;
3. an exact Bayesian quantum filter for weak localization on a nine-state nonorthogonal qutrit phase orbit;
4. a separation between historical-label goals and present-state goals;
5. a concrete continuous-control predictive-atlas research program.

2 Quantum and reinforcement-learning background

2.1 Quantum instruments

A state on $\mathcal{H} \cong \mathbb{C}^d$ is a positive semidefinite matrix ρ with $\text{tr } \rho = 1$. For action a and observed outcome o , a quantum instrument has completely positive branches

$$\mathcal{E}_o^a(\rho) = \sum_k K_{o,k}^a \rho K_{o,k}^{a\dagger}, \quad \sum_{o,k} K_{o,k}^{a\dagger} K_{o,k}^a = I. \quad (2)$$

The probability and conditional state are

$$p(o \mid \rho, a) = \text{tr } \mathcal{E}_o^a(\rho), \quad \rho' = \frac{\mathcal{E}_o^a(\rho)}{p(o \mid \rho, a)}. \quad (3)$$

Thus a measurement both informs and disturbs. This is the key feature absent from a classical likelihood table [6, 7, 8].

A random-unitary retry instrument is

$$K_s = \sqrt{p} U, \quad K_f = \sqrt{1-p} I. \quad (4)$$

Its success probability is state independent; failure leaves the state unchanged; success applies U .

2.2 Bellman optimality

In a finite Markov decision process, the shortest-path value for goal g obeys

$$V_g(g) = 0, \quad V_g(s) = \min_a \left[c(s, a) + \sum_{s'} P(s' \mid s, a) V_g(s') \right]. \quad (5)$$

The equality says that an optimal plan is consistent under one-step decomposition. When the hidden quantum state is not observed, a policy depends on the action–outcome history. A Bayesian belief or learned recurrent state must retain distinctions that matter to future outcome distributions and goals. Predictive-state representations formalize this operational criterion [5].

2.3 Sequence goals and memory

A sequence goal is a language over action–outcome symbols. A finite automaton updates $q_{t+1} = \delta(q_t, a_t, o_t)$; the Markov state is then (ρ_t, q_t) . This allows more goals than orthogonal quantum states, but the automaton memory is an additional resource. An exact unbounded $\mathbb{Z}^2$ displacement language needs counters or an episode horizon.

2.4 Metric and Euclidean tests

An optimal cost is an ordinary metric only if it is symmetric, positive, and satisfies the triangle inequality. For Euclidean compatibility, define

$$J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top, \quad B = -\frac{1}{2} J D^{\circ 2} J. \quad (6)$$

Schoenberg's theorem says that D embeds exactly in Euclidean space iff $B \succeq 0$, and in $\mathbb{R}^k$ iff additionally $\text{rank } B \leq k$ [12, 14]. Metric MDS seeks coordinates minimizing

$$\text{Stress}_k = \sqrt{\frac{\sum_{i < j} (\|x_i - x_j\|_2 - D_{ij})^2}{\sum_{i < j} D_{ij}^2}}. \quad (7)$$

Low stress is useful but does not replace the exact spectral test [13]. Spatial dynamics also require local, translation-consistent action increments, not merely a pleasing embedding.

3 Translation-covariant qutrit controls

3.1 Equilateral phase encoding

Define reciprocal vectors

$$q_0 = (1, 0), \quad q_1 = (-1/2, \sqrt{3}/2), \quad q_2 = (-1/2, -\sqrt{3}/2). \quad (8)$$

They obey

$$\sum_n q_n = 0, \quad \frac{1}{3} \sum_n q_n q_n^\top = \frac{1}{2} I_2.$$

Encode $x \in \mathbb{R}^2$ as

$$|\psi(x)\rangle = \frac{1}{\sqrt{3}} \sum_{n=0}^2 e^{i\kappa q_n \cdot x} |n\rangle, \quad \kappa = 0.15. \quad (9)$$

For displacement a ,

$$U_a = \text{diag} \left(e^{i\kappa q_0 \cdot a}, e^{i\kappa q_1 \cdot a}, e^{i\kappa q_2 \cdot a} \right) \quad (10)$$

satisfies $U_a |\psi(x)\rangle = |\psi(x+a)\rangle$ exactly. The same action translates every point; covariance is supplied by the group law rather than learned separately at each site.

3.2 Local projective geometry

The Fubini–Study distance is

$$d_{\text{FS}}(x, y) = \arccos |\langle \psi(x) | \psi(y) \rangle|.$$

Expansion around $dx = 0$, using the reciprocal-vector identities, gives

$$d_{\text{FS}}(x, x + dx)^2 = \frac{\kappa^2}{2} \|dx\|_2^2 + O(\|dx\|_2^4). \quad (11)$$

The qutrit orbit is locally isotropic. It is also compact and recurrent, so its ambient distance cannot equal an unbounded homogeneous history-space distance globally [11].

3.3 Local Kraus attempts

For each allowed stencil vector a , use

$$K_{a,s} = \sqrt{p_a} U_a, \quad K_{a,f} = \sqrt{1 - p_a} I_3. \quad (12)$$

Every attempt costs one. Parameters are shared under sign changes and coordinate permutations. Optimization can choose a cost for each primitive symmetry class but cannot introduce a goal- or site-specific action.

4 The finite-direction theorem

4.1 Exact reduction to a word metric

Let $S \subset \mathbb{Z}^2 \setminus \{0\}$ be finite, symmetric, and spanning. Give action v attempt cost $c_v > 0$ and success probability $p_v > 0$, and define $\ell_v = c_v/p_v$.

Proposition 4.1 (Exact local hodology). *If failure leaves the operational state unchanged and supplies no additional hidden-state information, then*

$$D_S(x, g) = \inf \left\{ \sum_{k=1}^n \ell_{v_k} : v_k \in S, \sum_{k=1}^n v_k = g - x \right\}. \quad (13)$$

Proof. Retrying v until success takes a geometric number of attempts with mean $1/p_v$, so every successful word is achievable at expected cost $\sum_k \ell_{v_k}$. Let V be the right side. Its triangle inequality gives $V(x) - V(x + v) \leq c_v/p_v$. Therefore

$$c_v + (1 - p_v)V(x) + p_v V(x + v) \geq V(x).$$

No first action or adaptive continuation can beat V . Positivity and reachability make it the minimal nonnegative Bellman solution. $\square$

4.2 Exact Euclidean rays

Proposition 4.2 (Ray criterion). *If $\ell_v = \|v\|_2$, then*

$$D_S(0, z) = \|z\|_2$$

iff z is a sum of available vectors all lying on its oriented ray.

Proof. For every word summing to z ,

$$\sum_k \|v_k\|_2 \geq \left\| \sum_k v_k \right\|_2 = \|z\|_2.$$

Strict convexity makes equality equivalent to all nonzero summands being nonnegative collinear vectors. $\square$

Corollary 4.3. *No fixed finite catalog is exactly Euclidean on all of $\mathbb{Z}^2$. Exactness on a radius- R square needs a primitive action ray for every coprime pair (a, b) with $0 \leq a, b \leq R$, hence $O(R^2)$ headings.*

4.3 Polygonal stable norm

Define

$$P_S = \text{conv}\{v/\ell_v : v \in S\}, \quad \|x\|_{P_S} = \inf\{\lambda > 0 : x \in \lambda P_S\}. \quad (14)$$

Theorem 4.4 (Stable local-control norm). *If S generates $\mathbb{Z}^2$, then*

$$\lim_{n \rightarrow \infty} \frac{D_S(0, nz)}{n} = \|z\|_{P_S}.$$

For finite S , this is a polygonal norm and cannot equal the Euclidean norm in every direction.

Proof. The gauge has the linear-program form

$$\|x\|_{P_S} = \inf_{\alpha_v \geq 0} \left\{ \sum_v \alpha_v \ell_v : \sum_v \alpha_v v = x \right\}.$$

Every integer word is feasible, giving the lower bound. Conversely, multiply an optimal fractional decomposition by n , round its coefficients, and correct the bounded lattice residue using a fixed generating subset. Thus

$$n \|z\|_{P_S} \leq D_S(0, nz) \leq n \|z\|_{P_S} + C_z.$$

Division by n proves convergence. A finite convex hull is a polygon, while the Euclidean unit disk has infinitely many extreme points. $\square$

This is a flat Finsler continuum limit [15, 16]. Refining the site spacing with fixed headings samples the polygon more densely but does not make it circular.

4.4 Angular error and exact examples

If $\ell_v = \|v\|_2$ and $\Delta_{\max}$ is the largest angular gap between adjacent headings, chord geometry gives

$$1 \leq \frac{\|x\|_{P_S}}{\|x\|_2} \leq \sec(\Delta_{\max}/2). \quad (15)$$

For K evenly spaced oriented directions,

$$\sec(\pi/K) - 1 = \frac{\pi^2}{2K^2} + O(K^{-4}).$$

Four axial moves give the Manhattan norm. Adding diagonals of cost $\sqrt{2}$ gives the octile norm

$$D_8(a, b) = A + (\sqrt{2} - 1)B, \quad A = \max(|a|, |b|), \quad B = \min(|a|, |b|). \quad (16)$$

Its asymptotic worst excess is 0.0823922; at $(2, 1)$ the excess is

$$\frac{1 + \sqrt{2}}{\sqrt{5}} - 1 = 0.0796691.$$

Adding all knight headings $(\pm 2, \pm 1)$ and permutations can make a 3×3 patch exact if their costs are Euclidean, but larger patches contain missing slopes. Small-patch exactness does not prove asymptotic Euclidean geometry.

Table 1: Fixed local repertoires. Locality ratio is maximum primitive step divided by held-out evaluation radius 12.

Name	directions	parameters	max. step	locality ratio
D4	4	1	1.000	0.083
D8	8	2	1.414	0.118
D16	16	3	2.236	0.186
D32	32	5	3.606	0.300

5 Control inverse design and evaluation

5.1 Repertoires

D4 has axes, D8 adds diagonals, D16 adds knight headings, and D32 adds primitive slopes with maximum coordinate three. The catalog is fixed before source and goal are supplied.

5.2 Targets, optimization, and held-out test

Translation covariance reduces planning to

$$V(0) = 0, \quad V(x) = \min_{a \in \mathcal{A}} \{c_a + V(x - a)\}, \quad (17)$$

solved exactly by Dijkstra’s algorithm. The targets are Euclidean distance and

$$\tilde{d}_{\text{FS}}(0, x) = \frac{\sqrt{2}}{\kappa} \arccos |\langle \psi(0) | \psi(x) \rangle|.$$

The latter agrees with Euclidean distance infinitesimally but contains compact projective curvature at finite radius.

A deterministic projected coordinate search minimizes mean squared relative error on all 48 nonzero integer displacements within radius four. Costs begin at $1.18 \|a\|_2$, satisfy $c_a \geq 1$, and share square symmetries. All 392 displacements with radii $4 < r \leq 12$ are held out. Evaluation reports quantum validity, Bellman residual, held-out errors, radial anisotropy, MDS stress, Schoenberg spectrum, and locality. These metrics prevent a persuasive plot from substituting for geometric evidence.

6 Control results

All validity checks close at numerical precision: maximum Kraus completeness residual 2.22×10^{-16} , covariance residual 1.67×10^{-16} , and Bellman residual 1.78×10^{-15} .

D8 is the best locality–accuracy compromise. Its diagonal expected cost is 1.32836, or success probability 0.75281. It is slightly cheaper than $\sqrt{2}$, balancing finite-disk relative errors rather than exactly matching the asymptotic circle.

D16 does not improve on D8. Its knight cost remains 2.63856, above a cheaper diagonal-plus-axial composite, so the added velocity lies inside the effective hull. Action count alone is not geometric expressivity; only exposed hull vertices matter. D32 activates a (3, 1) heading with cost 3.16229, close to $\sqrt{10}$, and achieves 1.62% held-out RMSE, at the cost of a longer maximum primitive.

At radius 12, directional range divided by mean is about 0.329 for D4, 0.110 for D8/D16, and 0.065 for D32. On the 25-goal diagnostic patch, Schoenberg negative-eigenmass falls from 0.2254 to

Table 2: Held-out performance on 392 displacements. Errors are relative to the named target.

Target	catalog	RMSE	mean error	max. abs. error
Euclidean	D4	0.3003	0.2719	0.4142
Euclidean	D8	0.0385	0.0220	0.0607
Euclidean	D16	0.0385	0.0220	0.0607
Euclidean	D32	0.0162	0.0022	0.0491
scaled FS	D4	0.3337	0.3063	0.4954
scaled FS	D8	0.0614	0.0484	0.1240
scaled FS	D16	0.0614	0.0484	0.1240
scaled FS	D32	0.0365	0.0276	0.0898

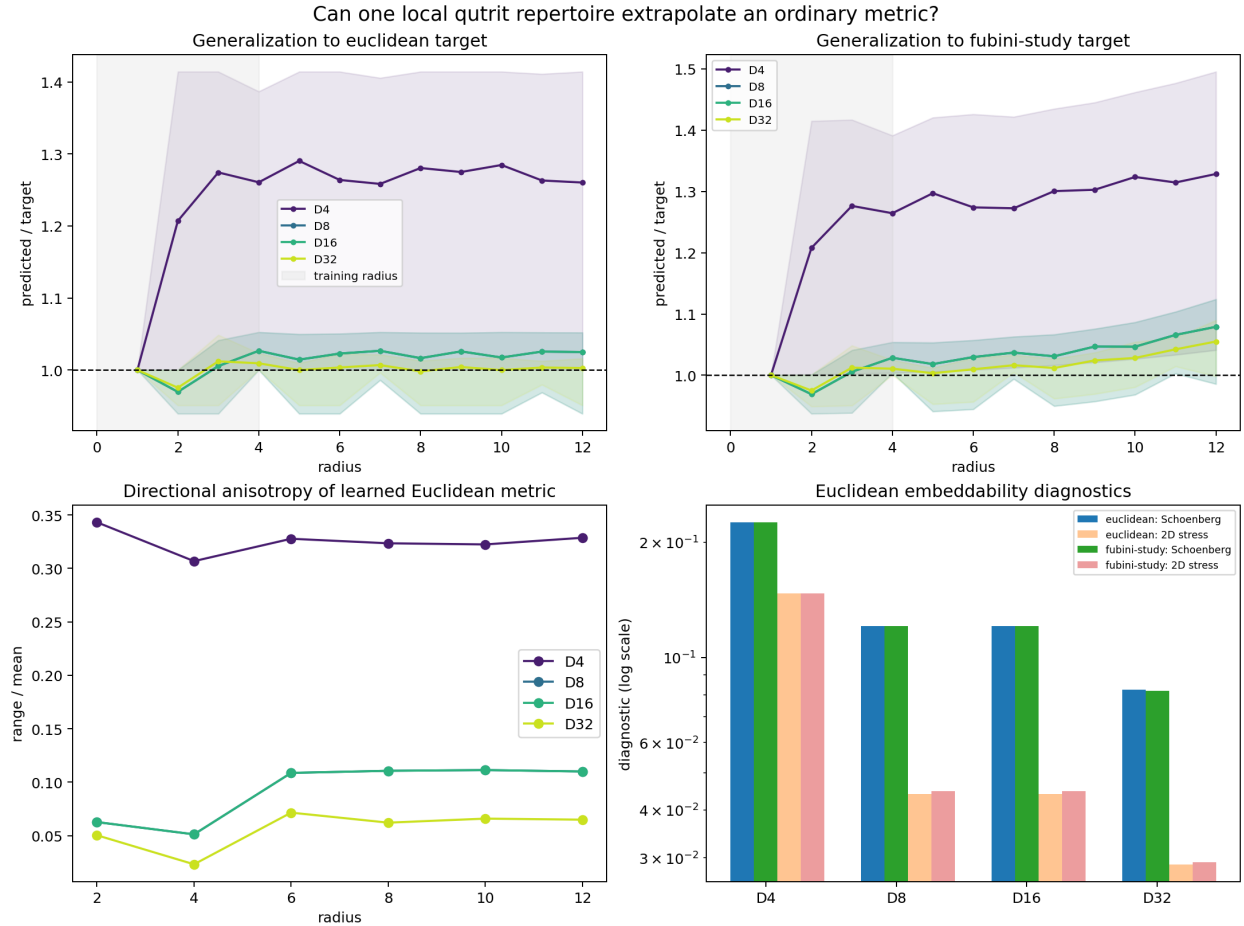


Figure 1: Local-control results. Top: predicted-to-target ratios by radius for Euclidean and scaled Fubini–Study targets. The gray region is fitted; larger radii are held out. Shading is directional range, not uncertainty. Bottom: anisotropy and Euclidean embeddability. More exposed directions improve the geometry but do not make a finite catalog exactly Euclidean.

0.1207 and 0.08236; 2D MDS stress falls from 0.1469 to 0.04409 and 0.02887. D32 looks convincingly planar, yet the spectral test correctly refuses an exact Euclidean interpretation.

7 Hidden-start localization

7.1 Why geometry alone is insufficient

The control study assumes a known source, so the agent can integrate observed successes. It tests the base metric but not access to position. A second model therefore hides a uniformly sampled initial label and provides weak quantum measurements. A finite $\mathbb{Z}_3^2$ orbit permits exact filtering and analytic controls.

Let $\omega = e^{2\pi i/3}$ and

$$|\phi_{xy}\rangle = \frac{|0\rangle + \omega^x|1\rangle + \omega^y|2\rangle}{\sqrt{3}}, \quad (x, y) \in \mathbb{Z}_3^2. \quad (18)$$

These nine states are not a basis. Distinct pairs have fidelity zero or $1/3$, and

$$\frac{1}{3} \sum_{x,y} |\phi_{xy}\rangle\langle\phi_{xy}| = I_3. \quad (19)$$

The commuting unitaries

$$U = \text{diag}(1, \omega, 1), \quad V = \text{diag}(1, 1, \omega)$$

translate the two labels modulo three. The moves $U, U^\dagger, V, V^\dagger$ have unit cost and give a toroidal Manhattan metric. This model studies localization, not the Euclidean stencil fit.

7.2 Weak covariant sensor

For strength $0 \leq \eta \leq 1$, outcome $o \in \mathbb{Z}_3^2$ has effect and Lüders Kraus operator

$$E_o(\eta) = \frac{\eta}{3} |\phi_o\rangle\langle\phi_o| + \frac{1-\eta}{9} I_3, \quad K_o(\eta) = \sqrt{E_o(\eta)}. \quad (20)$$

The frame identity gives $\sum_o E_o = I$. At $\eta = 0$, outcomes are uniform noise and the normalized state is unchanged. At $\eta = 1$, the rank-one phase POVM resets the state to the reported $|\phi_o\rangle$. Intermediate strength trades preparation information against disturbance [9, 10, 8].

Each episode samples a label S uniformly and prepares $|\phi_S\rangle$. The label is a hidden laboratory record used to score ensemble discrimination, not an ontic property of the average state:

$$\bar{\rho}_0 = \frac{1}{9} \sum_s |\phi_s\rangle\langle\phi_s| = \frac{I}{3}.$$

The same mixed state admits many ensemble decompositions.

7.3 Exact Bayesian quantum filter

A classical likelihood table is wrong after the first observation because backaction changes every hidden-preparation branch. The exact filter stores a posterior weight w_s and conditional state ρ_s for each hypothesis. For observed o ,

$$\ell_s(o) = \text{tr}(E_o \rho_s), \quad p(o) = \sum_s w_s \ell_s(o), \quad (21)$$

$$w'_s = \frac{w_s \ell_s(o)}{p(o)}, \quad \rho'_s = \frac{K_o \rho_s K_o^\dagger}{\ell_s(o)}. \quad (22)$$

The present predictive state is

$$\bar{\rho} = \sum_s w_s \rho_s. \quad (23)$$

This nine-branch filter is the exact benchmark for a future learned recurrent predictive representation.

7.4 Two goals on the same history

The experiment scores two criteria.

1. **Preparation-label navigation.** Let $\hat{S} = \arg \max_s w_s$. Apply the translation that would move $\hat{S}$ to goal g . Label success means $\hat{S} = S$.
2. **Present-state navigation.** Find the phase-grid projector with greatest fidelity to $\bar{\rho}$, translate that center to g , and score $\langle \phi_g | \rho_{\text{final}} | \phi_g \rangle$.

The first asks where the episode began; the second asks which action controls the state that exists now. They coincide in a perfectly observed nondisturbing classical register, but not here.

Proposition 7.1 (One-shot information–preparation gap). *With a uniform preparation and one sensor outcome,*

$$P(\hat{S} = S) = \frac{1 + 2\eta}{9}, \quad F_g = \frac{1 + 2\eta}{3}, \quad (24)$$

where F_g is the final fidelity obtained by translating the reported outcome to g .

Proof. Since the predictive state is $I/3$, every outcome has $p(o) = \text{tr}(E_o I/3) = 1/9$. The matching hypothesis has likelihood

$$p(o | s = o) = \langle \phi_o | E_o | \phi_o \rangle = \frac{\eta}{3} + \frac{1 - \eta}{9} = \frac{1 + 2\eta}{9}.$$

Symmetry makes it the MAP hypothesis, proving the first formula. The conditioned ensemble state is

$$\bar{\rho}_o = \frac{K_o(I/3)K_o}{1/9} = 3E_o = \eta|\phi_o\rangle\langle\phi_o| + (1 - \eta)\frac{I}{3}.$$

Translating o to g yields fidelity $\eta + (1 - \eta)/3 = (1 + 2\eta)/3$. □

At $\eta = 1$, initial-label accuracy is only $1/3$, while present-state fidelity is one. The first sharp outcome erases all remaining quantum dependence on the preparation. Later sharp outcomes contain no more information about S . Posterior entropy falls from $\log_2 9 = 3.169925$ bits to 2.641604 bits and then remains fixed.

7.5 Policies and controls

Fixed rules use 0, 1, 3, or 5 observations. Adaptive rules stop when $\max_s w_s$ reaches 0.25, 0.32, or 0.50, with a six-sense cap. Every sensor, move, and terminal commitment costs one. Covariance makes translation before sensing a mere permutation of outcomes, so this first model studies stopping rather than probe choice.

The null sensor $\eta = 0$ changes neither posterior nor state; hidden-label success must remain $1/9$. The known-start control uses no sensor and reaches both terminal criteria exactly. Its uniform mean torus movement distance is $4/3$, so mean total cost including commitment is $7/3$.

8 Localization results

8.1 Protocol and representative results

The production bundle contains 60,000 seeded episodes: five strengths, seven sensing rules plus a known-start control, and 1,500 episodes per condition. Eight tests cover POVM and Kraus completeness, covariance, null sensing, exact one-shot formulas, physical filter updates, known-start navigation, and reproducibility. The maximum discrepancy from the analytic one-shot formulas over all strengths is 0.0196.

At weak and medium strength, repeated observations accumulate origin information. At $\eta = 0.5$, label success rises from 0.203 after one observation to 0.271 after five, but total cost rises from 3.34 to 7.36. The threshold-0.25 policy uses 4.16 observations, retains label success 0.264, and reaches label-policy fidelity 0.807, better and cheaper than fixed five.

At high strength, origin inference and present-state control diverge. At $\eta = 0.8$, five observations raise label accuracy from 0.273 to 0.302, yet historical-label fidelity falls from 0.870 to 0.510. The predictive-state controller reaches 0.901. At $\eta = 1$, repeated sensing cannot improve the origin posterior, but the predictive-state controller retains unit fidelity by following the state prepared by the most recent measurement.

Null sensing stays at chance and adds cost. Known-start motion is exact. Informative sensing helps retrodiction but is limited by nonorthogonality and backaction. These controls show that failure is neither broken movement nor blind homing, and that measurement-induced preparation is not synonymous with localization.

9 Joint geometric interpretation

9.1 A Finsler base

The control experiment identifies a homogeneous base action geometry whose exact large-scale object is $\|\cdot\|_{P_S}$. D4 is Manhattan; D8 and D32 are increasingly circular polygons. The base is spatial in locality, translation covariance, and low-distortion planar embedding, but it is not exactly Riemannian for a finite heading set.

The learned geometry is not ambient qutrit-ray distance. It is the Bellman geometry of available transformations. Compact Hilbert-space recurrence can coexist with an unbounded history-space covering coordinate.

9.2 An epistemic quantum fiber

With a hidden source, one coordinate is insufficient. Above a nominal phase site lie posterior weights, hypothesis-conditioned density matrices, the present ensemble state, and possibly sequence-goal automaton state. Measurement transports this internal structure and changes which base point best predicts future control. This resembles

$$F_x \hookrightarrow E \xrightarrow{\pi} B,$$

with movement geometry B and an epistemic/internal fiber F_x . It is not yet a mathematical fiber bundle: local trivializations, transition functions, and a connection have not been established. It is an operational precursor with measurable base–fiber coupling.

Table 3: Representative localization outcomes. Label-policy fidelity moves according to the origin MAP estimate; state-policy fidelity moves according to the present predictive state.

η	senses	rule	label success	label fidelity	state fidelity	mean senses	total cost
0.0	0	none	0.105	0.328	0.328	0.00	1.00
0.0	5	fixed	0.099	0.325	0.325	5.00	6.00
0.2	1	fixed	0.147	0.459	0.459	1.00	3.33
0.2	5	fixed	0.189	0.548	0.553	5.00	7.32
0.5	1	fixed	0.203	0.659	0.659	1.00	3.34
0.5	5	fixed	0.271	0.735	0.786	5.00	7.36
0.5	—	adaptive 0.25	0.264	0.807	0.831	4.16	6.50
0.8	1	fixed	0.273	0.870	0.870	1.00	3.32
0.8	5	fixed	0.302	0.510	0.901	5.00	7.32
1.0	1	fixed	0.324	1.000	1.000	1.00	3.34
1.0	5	fixed	0.333	0.356	1.000	5.00	7.31
any	0	known start	1.000	1.000	1.000	0.00	≈ 2.33

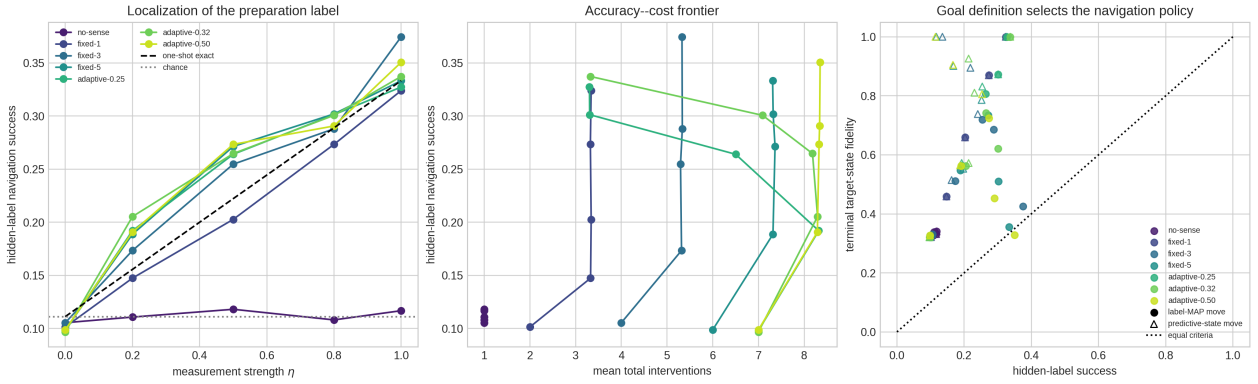


Figure 2: Localization and control. Left: preparation-label success versus strength. The dotted line is chance and the dashed line is the exact one-shot formula. Center: label accuracy versus total cost. Right: terminal fidelity versus label success. Filled circles use the historical-label move; hollow triangles use the present-state move.

9.3 Goal relativity

The origin-MAP controller and predictive-state controller answer different questions. If the goal is retrodiction, a newly prepared state is not success. If the goal is present control, insisting on the origin can be harmful. Every reported distance must specify the terminal criterion, cost convention, available information, and whether it concerns history, current predictive state, or a future outcome sequence. Their agreement in an orthogonal classical position register is a special limit.

10 Limitations

1. **Engineered primitive costs.** Goal-sized macro-actions are gone, but primitive probabilities are still optimized against target geometry. The study shows compositional extrapolation, not spontaneous selection of the Euclidean norm.
2. **Finite headings.** The polygonal theorem guarantees a nonzero anisotropy floor. D32 improves accuracy partly by extending primitive reach to 30% of the evaluation radius.
3. **Two qutrit models.** Control uses a continuous equilateral phase encoding; localization uses a finite $\mathbb{Z}_3^2$ orbit. They share phase translation structure but are not yet one end-to-end environment.
4. **Preparation label.** The hidden origin is a laboratory ensemble record, not an intrinsic observable property of $I/3$.
5. **Known filter.** The Bayesian model knows the instruments. A recurrent agent has not learned the filter from trajectories.
6. **Limited active sensing.** One covariant sensor family makes active design a stopping problem; inequivalent probes are absent.
7. **Statistics.** Control is deterministic. Localization is a single recorded seed with 1,500 episodes per row, analytic checks, and reported standard errors; multi-seed replication remains useful.
8. **No curvature yet.** The environments are homogeneous and periodic or evaluated far from recurrence. There is no spatially varying control tensor or demonstrated holonomy.

11 Continuous control and a predictive atlas

11.1 Replace the polygon by a disk

Let a qutrit have commuting phase generators A, B and continuous control

$$H(u) = u_1 A + u_2 B, \quad \dot{x} = u. \quad (25)$$

If admissible velocities obey $\|u\|_2 \leq 1$ and cost is elapsed time, the velocity body is a disk and $T^*(x, g) = \|g - x\|_2$. If instead $|u_i| \leq 1$, it is a square and minimum time is Chebyshev. Euclidean geometry is selected by isotropy of the resource body, not merely by two coordinates.

11.2 Quadratic effort and a metric tensor

For fixed duration T , define

$$\mathcal{J}[u] = \frac{1}{2} \int_0^T u(t)^\top R(x_t) u(t) dt. \quad (26)$$

For constant $R \succ 0$, Cauchy–Schwarz gives

$$\mathcal{J}^*(x, g; T) = \frac{1}{2T} (g - x)^\top R (g - x), \quad (27)$$

attained by constant velocity. The square root of $2T\mathcal{J}^*$ is a flat Riemannian distance. Position-dependent $R(x)$ defines a curved metric and a Hamilton–Jacobi control problem [18]. This replaces hand-tuned retries with a physical effort principle.

11.3 Learn the predictive state

The exact filter supplies scientific ground truth. A recurrent atlas should receive only (a_t, o_t) and learn

$$z_{t+1} = F_\theta(z_t, a_t, o_t)$$

that predicts next outcomes under contemplated interventions, goal-conditioned cost, terminal fidelity, and calibrated operational uncertainty. Auxiliary predictive likelihood is essential: a visually organized latent state is not enough. Compare navigation regret, posterior calibration, held-out likelihood, and latent probes directly with the exact filter.

11.4 Genuine active localization

Add rotated or coarse-grained phase POVMs so expected information gain depends on the current posterior. The agent must choose whether to sense, which probe to use, and measurement strength while accounting for disturbance. Small exact belief-space dynamic programs provide oracle controls for learned value-of-information policies.

11.5 Base, fiber, and curvature tests

Movement, sensing, and commitment should remain separate cost matrices. Movement estimates the base; sensing estimates epistemic overhead. A total cost can be useful for competence yet nonmetric. Calling its distortion curvature would confound movement with information.

Once a stable base chart and learned internal state exist, transport the same internal preparation around closed base loops. A reproducible loop-dependent internal transformation, after controlling for base displacement and sensor schedule, defines operational holonomy. Variation across nearby loops is the first credible curvature observable. This is the appropriate bridge toward bundle geometry, long before claims about general relativity.

11.6 Falsifiers

The next stage should be revised if rotational error does not decrease with a continuous control disk; a recurrent latent cannot match exact posterior calibration and predictive likelihood; apparent curvature vanishes when sensing and movement costs are separated; loop transport has no reproducible holonomy; or the geometry resides mainly in a hand-coded goal automaton rather than outcome statistics.

12 Conclusion

Low Hilbert dimension is not the decisive obstruction to a large spatial goal geometry. A qutrit supports many nonorthogonal phase goals, exact local translations, and partial localization. The decisive geometric constraint is the local control resource. A fixed finite heading catalog produces a polygonal stable norm. It can approximate a plane impressively—D32 reaches 1.62% held-out RMSE—but cannot become exactly rotationally invariant.

The localization study adds a second qualification. The place from which an episode began and the move that controls the state now are different quantum questions. Sharp sensing can prepare a perfectly controllable target while retaining only 1/3 origin accuracy. Operational state therefore contains a base coordinate and an internal posterior/quantum structure.

The strongest next model combines continuous isotropic qutrit control with a learned predictive atlas. Position-dependent control effort and closed-loop internal transport can then test metric curvature and connection. The present proofs, controls, and calibrated simulations provide the baseline required to distinguish genuine emergence from geometry inserted by hand.

A Stable-norm duality and active actions

The gauge has the linear program

$$\|x\|_{P_S} = \inf_{\alpha_v \geq 0} \left\{ \sum_v \alpha_v \ell_v : \sum_v \alpha_v v = x \right\}$$

and dual

$$\|x\|_{P_S} = \sup_{\xi \in \mathbb{R}^2} \{ \xi \cdot x : \xi \cdot v \leq \ell_v \text{ for all } v \}.$$

An action affects the asymptotic metric only if v/ℓ_v is an exposed point of P_S . This makes the D16 failure predictable: an action inside the D8 hull cannot change the gauge. Randomized policies fill P_S by convex mixtures but cannot curve its boundary.

General noisy increments can add diffusion corrections. An isotropic heat kernel may define a Riemannian metric through a Varadhan-type relation

$$d_g(x, y)^2 = \lim_{t \downarrow 0} -4t \log p_t(x, y)$$

[17]. This is not expected hitting time. In two dimensions an unbiased infinite random walk is recurrent but has infinite mean point-hitting time. Diffusion geometry is promising, but changes the operational definition and must be studied explicitly.

B Statistical formulas

For N primitive surveys,

$$\text{SE}(\hat{p}) \approx \sqrt{\frac{p(1-p)}{N}}.$$

With $\ell = 1/p$, the delta method gives

$$\frac{\text{SE}(\hat{\ell})}{\ell} \approx \sqrt{\frac{1-p}{Np}}.$$

Learned geometry should approach the exact polygonal model at $N^{-1/2}$, not cross its structural anisotropy floor. Binomial navigation standard errors use $\sqrt{\hat{q}(1-\hat{q})/N}$; fidelity standard errors use sample standard deviation divided by $\sqrt{N}$.

C Reproducibility

From the repository root:

```
python -m unittest discover \
    -s low_dimensional_hodology/emergent_local_metric/control/tests -v
MPLBACKEND=Agg python \
    low_dimensional_hodology/emergent_local_metric/control/run_local_control.py

python -m unittest discover \
    -s low_dimensional_hodology/emergent_local_metric/localization \
    -p 'test_*.py' -v
MPLBACKEND=Agg python \
    low_dimensional_hodology/emergent_local_metric/localization/\
    localization_experiment.py --episodes 1500 --seed 20260811
```

Control artifacts include optimized costs, held-out displacements, anisotropy, diagnostics, and the summary figure. Localization artifacts include full condition summaries, posterior calibration, exact one-shot predictions, representative episodes, a manifest, and both figures. The two strands contain 15 focused tests.

References

- [1] R. Bellman, *Dynamic Programming*, Princeton University Press (1957).
- [2] M. L. Puterman, *Markov Decision Processes*, Wiley (1994).
- [3] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed., MIT Press (2018).
- [4] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, Planning and acting in partially observable stochastic domains, *Artificial Intelligence* **101**, 99–134 (1998).
- [5] M. L. Littman, R. S. Sutton, and S. Singh, Predictive representations of state, in *Advances in Neural Information Processing Systems 14*, 1555–1561 (2002).
- [6] K. Kraus, *States, Effects, and Operations*, Springer (1983).
- [7] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).
- [8] H. M. Wiseman and G. J. Milburn, *Quantum Measurement and Control*, Cambridge University Press (2010).
- [9] A. S. Holevo, *Probabilistic and Statistical Aspects of Quantum Theory*, 2nd ed. (2011).
- [10] C. W. Helstrom, *Quantum Detection and Estimation Theory*, Academic Press (1976).

- [11] I. Bengtsson and K. Życzkowski, *Geometry of Quantum States*, 2nd ed., Cambridge University Press (2017).
- [12] I. J. Schoenberg, Remarks to Maurice Fréchet's article on Hilbert-space distance, *Annals of Mathematics* **36**, 724–732 (1935).
- [13] J. B. Kruskal, Multidimensional scaling by optimizing goodness of fit, *Psychometrika* **29**, 1–27 (1964).
- [14] I. Borg and P. J. F. Groenen, *Modern Multidimensional Scaling*, 2nd ed., Springer (2005).
- [15] D. Bao, S.-S. Chern, and Z. Shen, *An Introduction to Riemann–Finsler Geometry*, Springer (2000).
- [16] D. Burago, Y. Burago, and S. Ivanov, *A Course in Metric Geometry*, AMS (2001).
- [17] S. R. S. Varadhan, Diffusion processes in a small time interval, *Communications on Pure and Applied Mathematics* **20**, 659–685 (1967).
- [18] D. D'Alessandro, *Introduction to Quantum Control and Dynamics*, CRC Press (2008).

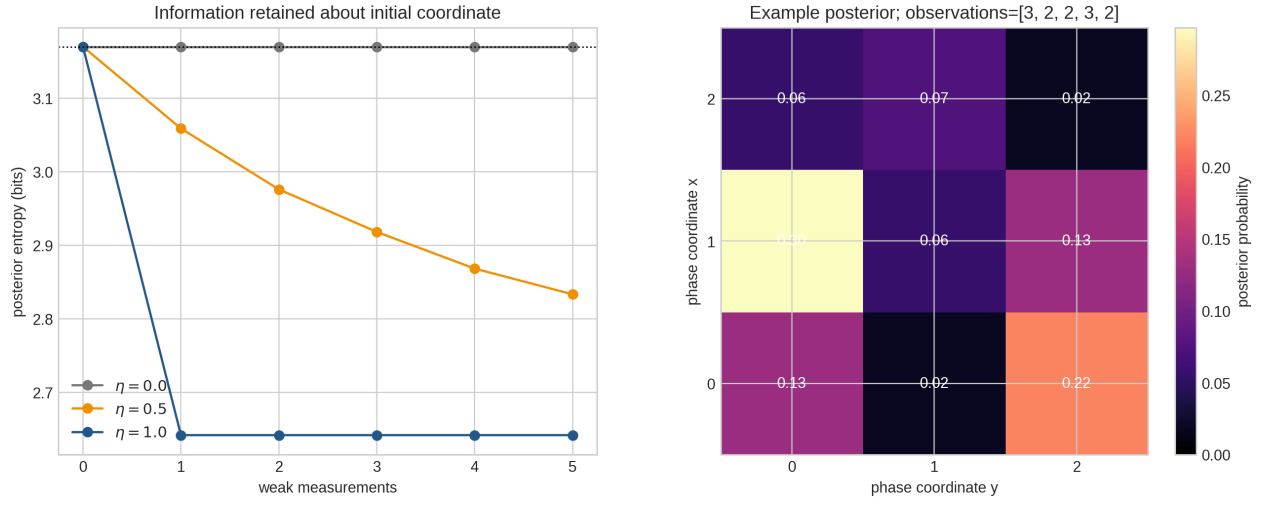


Figure 3: Belief dynamics. Left: posterior entropy about the initial coordinate. A sharp measurement extracts all retained origin information in one step and then saturates; intermediate strength accumulates information. Right: an example posterior after five outcomes. A point estimate cannot represent the full operational state.